List R1 — Introduction, accounting revision and the capital structure

Worked solutions

Finanças II — Corporativas (MPE Insper)

Statements are in `Listas/lista-r1-accounting-capital-structure.md`; each exercise is cited by number and printed page and is not restated. Editions are the local copies: Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013). Ross prints thousands with a period ($196.000); below they are written 196,000. Money is in the units of each statement.

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1. Transactions and the balance sheet — Higgins 1.4, p. 33

What is asked. For each of eight transactions, say which balance sheet accounts move, by how much and in which direction, and check that assets still equal liabilities plus equity afterwards. (a) selling used equipment above its book value; (b) buying a building with part cash, part bank loan;

(c) buying a building with cash only; (d) paying suppliers; (e) buying back own shares; (f) a cash sale of merchandise; (g) a credit sale of merchandise; (h) paying a dividend.

Approach. Every transaction is recorded twice, so the balance sheet identity $A = L + E$ must hold in changes: $\Delta A = \Delta L + \Delta E$. For each item, write the two (or three) accounts that move, then read ΔE off the identity. Equity moves only when owners gain or lose value (a gain, a profit, a payout, a buyback); swaps of one asset for another do not touch it.

Notation.

- A, L, E: total assets, total liabilities, owners' equity (\$; $A = L + E$).
- ΔX: change in account X caused by the transaction (\$, any sign).
- C_f, C_g: book (inventory) cost of the merchandise sold in (f) and (g) (\$, ≥ 0; not given).

Assumption. Taxes are ignored (a gain or profit goes to retained earnings in full); in (e) the repurchased shares are carried as treasury stock, a deduction from equity.

Step 1 — The identity in changes. Subtracting the identity before the transaction from the identity after it gives the rule used in every part:

$$A = L + E, \tag{1}$$

$$A + \Delta A = (L + \Delta L) + (E + \Delta E),$$

$$\Delta A = \Delta L + \Delta E. \tag{2}$$

Step 2 — (a) Sale of used equipment. Cash rises by the price received and the equipment account falls by its book value, so total assets change by the difference:

$$\begin{aligned} \Delta A &= \Delta\text{Cash} + \Delta\text{Equipment} \\ &= (500{,}000) + (-300{,}000) \\ &= 200{,}000. \end{aligned} \tag{3}$$

No liability changes, $\Delta L = 0$, so (2) gives $\Delta E = \Delta A - \Delta L = 200{,}000 - 0 = 200{,}000$: the gain on sale (price minus book value) enters retained earnings.

Answer (a). Cash +\$500,000, equipment −\$300,000; assets and equity (retained earnings) both rise by \$200,000.

Step 3 — (b) Building financed 40% cash, 60% loan. The cash part and the loan part are each a share of the price:

$$\begin{aligned} \Delta\text{Cash} &= -(0.40)(80) \\ &= -32 \text{ million}, \end{aligned} \tag{4}$$

$$\begin{aligned} \Delta\text{Bank loan} &= (0.60)(80) \\ &= 48 \text{ million}. \end{aligned} \tag{5}$$

Assets gain the building and lose the cash:

$$\begin{aligned} \Delta A &= \Delta\text{Buildings} + \Delta\text{Cash} \\ &= (80) + (-32) \\ &= 48 \text{ million}. \end{aligned}$$

By (2), $\Delta E = \Delta A - \Delta L = 48 - 48 = 0$.

Answer (b). Buildings +\$80 million, cash −\$32 million, bank loan +\$48 million; assets and liabilities rise by \$48 million, equity unchanged.

Step 4 — (c) Building bought for cash. One asset is exchanged for another of equal value:

$$\begin{aligned}\Delta A &= \Delta\text{Buildings} + \Delta\text{Cash}\\ &= (60) + (-60)\\ &= 0.\end{aligned}$$

With $\Delta L = 0$, (2) gives $\Delta E = 0$. Only the composition of assets moves (from liquid cash to an illiquid building).

Answer (c). Buildings +\$60 million, cash −\$60 million; totals unchanged.

Step 5 — (d) Payment to trade creditors. Paying a supplier settles a liability with cash: $\Delta\text{Cash} = -40{,}000$ and $\Delta\text{Accounts payable} = -40{,}000$, so $\Delta A = \Delta L = -40{,}000$ and, by (2), $\Delta E = -40{,}000 - (-40{,}000) = 0$.

Answer (d). Cash and accounts payable both fall by \$40,000; equity unchanged.

Step 6 — (e) Share repurchase. The firm pays the price on each share bought back:

$$\begin{aligned}\Delta\text{Cash} &= -(10{,}000)(24)\\ &= -240{,}000.\end{aligned} \tag{6}$$

No liability changes, so by (2) $\Delta E = \Delta A = -240{,}000$ (recorded as treasury stock, a deduction from equity).

Answer (e). Cash −\$240,000 and equity −\$240,000; liabilities unchanged.

Step 7 — (f) Cash sale of merchandise. Cash comes in at the sale price and inventory leaves at its cost: $\Delta A = \Delta\text{Cash} + \Delta\text{Inventory} = 80{,}000 - C_f$. With $\Delta L = 0$, (2) gives $\Delta E = 80{,}000 - C_f$, the gross profit, added to retained earnings.

Answer (f). Cash +\$80,000, inventory $-C_f$; equity rises by the gross profit $80{,}000 - C_f$.

Step 8 — (g) Credit sale of merchandise. The same as (f), except the asset received is a claim on the customer: $\Delta A = \Delta\text{Receivables} + \Delta\text{Inventory} = 120{,}000 - C_g$, so $\Delta E = 120{,}000 - C_g$.

Answer (g). Accounts receivable +\$120,000, inventory $-C_g$; equity rises by $120{,}000 - C_g$.

Step 9 — (h) Dividend. A dividend pays cash out of retained earnings: $\Delta\text{Cash} = -50{,}000$, $\Delta L = 0$, so $\Delta E = \Delta A = -50{,}000$.

Answer (h). Cash and retained earnings both fall by \$50,000.

Intuition. Every transaction has two sides. Only exchanges with outsiders that create or destroy value for owners (a gain, a profit, a payout, a buyback) move equity; swaps of one asset for another, or borrowing to buy an asset, change the composition or size of the balance sheet without touching equity. This is why book equity is a residual: it is whatever is left after liabilities are subtracted from assets.

Common mistakes.

- Booking the full \$500,000 in (a) as a gain: the gain is only the excess over book value, \$200,000.
- Treating a credit sale (g) as having no effect until cash arrives: accrual accounting records it at the sale.
- Recording a dividend (h) or a buyback (e) as an expense: both are distributions to owners and bypass the income statement.

Ref: Higgins Ch. 1, § "The Balance Sheet" (with "Current Assets and Liabilities"), pp. 9–11; § "Measuring Earnings", p. 12. No printed answer (even-numbered problem).

2. Local Co. margins — Berk 2.12, p. 54

What is asked. (a) The share of each sales dollar left after the direct cost of the goods sold (gross margin). (b) The share left after also paying the costs of running the business, including depreciation (operating margin). (c) The share left for shareholders after taxes (net profit margin).

Approach. Build the income statement top-down (Berk's Table 2.3 layout) and divide each subtotal by sales. Each margin is one more layer of cost removed: production, operations, then financing and taxes (Berk Eqs. 2.9–2.11).

Notation.

- S: sales; C: cost of sales; SGA: selling, general and administrative expenses; RD: research and development; D: depreciation ($ million, ≥ 0).
- τ: tax rate ($0 \leq \tau < 1$); GP: gross profit; OI: operating income; NI: net income.

Assumption. No interest expense and no other income (none is given), so OI is also pretax income. Depreciation is an operating expense, as in Berk's Table 2.3 (p. 34). The $500,000 of SG&A is 0.5 million.

Step 1 — Gross profit and gross margin (a). Gross profit removes only the cost of the goods sold; dividing by sales gives the margin:

$$\begin{aligned} GP &= S - C \\ &= (10) - (6) \\ &= 4, \end{aligned} \tag{7}$$

$$\begin{aligned} \text{Gross margin} &= GP/S \\ &= (4)/(10) \\ &= 0.40 \\ &= 40\%. \end{aligned} \tag{8}$$

Answer (a). Gross margin = 40%.

Step 2 — Operating income and operating margin (b). Operating income subtracts every operating expense from gross profit (7):

$$\begin{aligned} OI &= GP - SGA - RD - D \\ &= (4) - (0.5) - (1) - (1) \\ &= 1.5, \end{aligned} \tag{9}$$

$$\begin{aligned} \text{Operating margin} &= OI/S \\ &= (1.5)/(10) \\ &= 0.15 \\ &= 15\%. \end{aligned}$$

Answer (b). Operating margin = 15%.

Step 3 — Net income and net profit margin (c). With no interest, pretax income is OI from

(9); taxes are τ times it, and net income is what remains:

$$\begin{aligned} Tax &= \tau\, OI \\ &= (0.35)(1.5) \\ &= 0.525, \\ NI &= OI - Tax \\ &= (1.5) - (0.525) \\ &= 0.975, \\ \text{Net profit margin} &= NI/S \\ &= (0.975)/(10) \\ &= 0.0975 \\ &= 9.75\%. \end{aligned} \tag{10}$$

Answer (c). Net profit margin = 9.75%.

Intuition. Each margin peels off one more layer of cost: production costs (gross), the cost of running the business (operating), then financing and taxes (net). The gap between operating and net margin here is only the 35% tax, because Local has no debt: $15\% \times (1 - 0.35) = 9.75\%$.

Common mistakes.

- Leaving depreciation out of operating expenses: it is non-cash, but it is an operating cost in the income statement.
- Mixing units: SG&A is \$500,000, i.e. 0.5 million, not 500 million.
- Applying the tax to gross profit instead of pretax income.

Ref: Berk Ch. 2, §2.4 "The Income Statement", pp. 33–35; §2.5 "Income Statement Analysis" (Eqs. 2.9–2.11), pp. 35–36.

3. Nokela: earnings versus cash flow — Berk 2.31, p. 57

What is asked. (a) By how much the purchase changes reported net income in each of the four years. (b) By how much it changes the firm's cash flow in each of the four years.

Approach. Earnings see the purchase only through depreciation, an expense spread over four years that is deductible for tax. Cash flow sees the purchase price when it is paid and, afterwards, only the tax that depreciation saves (the depreciation tax shield). Comparing the two shows that earnings and cash differ in timing, not in total.

Notation.

- $I = 40$: purchase price (\$ million), paid in year 1 ("this year").
- $D = 10$: straight-line depreciation in each year $t = 1, \ldots, 4$ (\$ million).
- $\tau = 0.40$: tax rate; ΔNI_t, ΔCF_t: change in earnings and in cash flow in year t.

Assumption. Nokela has enough other taxable income to use the depreciation deduction each year.

Step 1 — Impact on earnings (a). Depreciation lowers pretax income by D; taxes then fall by τD, so net income falls by the difference:

$$\begin{aligned} \Delta NI_t &= -D + \tau D \\ &= -D(1 - \tau) \\ &= -(10)(1 - 0.40) \\ &= -(10)(0.60) \\ &= -6 \quad \text{for each } t = 1, \ldots, 4. \end{aligned} \tag{11}$$

Answer (a). Earnings fall by \$6 million in each of the four years.

Step 2 — Impact on cash flow (b). The purchase is a cash outflow in year 1. Depreciation is not paid to anyone, but it lowers the tax bill by τD every year:

$$\begin{aligned} \tau D &= (0.40)(10) \\ &= 4, \qquad (12) \\ \Delta CF_1 &= -I + \tau D \\ &= -(40) + (4) \\ &= -36, \qquad (13) \\ \Delta CF_t &= \tau D = 4 \quad \text{for } t = 2, 3, 4. \qquad (14) \end{aligned}$$

Answer (b). Cash flow changes by −\$36 million in year 1 and +\$4 million in each of years 2–4.

Step 3 — Consistency check. Over the four years both measures add to the same total, using (11)–(14):

$$\begin{aligned} \textstyle\sum_t \Delta NI_t &= 4(-6) \\ &= -24, \\ \textstyle\sum_t \Delta CF_t &= (-36) + 3(4) \\ &= (-36) + (12) \\ &= -24. \end{aligned}$$

Both equal the after-tax cost of the machine, $-I(1-\tau) = -(40)(0.60) = -24$.

Intuition. Accounting spreads the cost evenly (−6 a year); cash flow records it when paid (−36 up front) and then gives back the tax shield (+4 a year). Same total, different timing, and timing is what matters for value: at any positive discount rate the cash-flow pattern is worse for the firm than the earnings pattern suggests.

Common mistakes.

- Charging the whole \$40 million against year-1 earnings: a capital expense reaches earnings only through depreciation.
- Treating depreciation as a cash outflow in (b): its only cash effect is the tax saving τD.
- Forgetting the tax effect in (a) and answering −10 a year.

Ref: Berk Ch. 2, §2.4, pp. 33–35 (depreciation as a non-cash expense); §2.6 "The Statement of Cash Flows", pp. 42–45.

4. GE: book versus market — Berk 2.8, p. 53

What is asked. (a) The market value of GE's equity, and how many times book equity it is. (b) Debt relative to equity, once with book equity and once with market equity. (c) The value of GE's operating business net of its cash (enterprise value).

Approach. Market capitalization is shares times price; ratios then follow from their definitions in Berk §2.3. Enterprise value (Berk Eq. 2.5) adds debt to market equity and subtracts cash, because a buyer of the whole firm would pay off the debt and keep the cash.

Notation.

- $N = 10.3$ billion shares; $P = 38$: share price (\$); $BE = 117$: book equity (\$ billion).
- $D = 467$: total debt; $Cash = 16$ (\$ billion); MV_E: market capitalization; EV: enterprise value.

Step 1 — Market capitalization and market-to-book (a). Multiply shares by price, then divide by book equity:

$$\begin{aligned} MV_E &= N \times P \\ &= (10.3)(38) \\ &= 391.4, \\ M/B &= MV_E/BE \\ &= (391.4)/(117) \\ &= 3.345. \end{aligned} \tag{15}$$

Answer (a). Market capitalization = \$391.4 billion; market-to-book = 3.35.

Step 2 — Debt-equity ratios (b). The same debt is divided by each measure of equity, book and then market from (15):

$$\begin{aligned} (D/E)_{\text{book}} &= D/BE \\ &= (467)/(117) \\ &= 3.991, \\ (D/E)_{\text{market}} &= D/MV_E \\ &= (467)/(391.4) \\ &= 1.193. \end{aligned}$$

Answer (b). Book $D/E = 3.99$; market $D/E = 1.19$.

Step 3 — Enterprise value (c). First net the cash against debt, then add market equity from (15) (Berk Eq. 2.5):

$$\begin{aligned} \text{Net debt} &= D - Cash \\ &= (467) - (16) \\ &= 451, \\ EV &= MV_E + \text{Net debt} \\ &= (391.4) + (451) \\ &= 842.4 \text{ billion}. \end{aligned}$$

Answer (c). Enterprise value = \$842.4 billion.

Intuition. Investors valued GE's equity at more than three times its book value, so leverage measured at book (4.0) overstates the claim of debt relative to what equity is worth (1.2). Enterprise value nets out cash: it is the value of the operating business that a buyer of all the equity and all the debt would acquire. For capital structure questions the market ratio is the relevant one, since it compares claims at the prices investors pay for them.

Common mistakes.

- Using book equity in enterprise value: EV uses market capitalization.
- Adding cash instead of subtracting it in EV.
- Mixing units (shares in billions, price in dollars, debt in billions): keep everything in billions of dollars.

Ref: Berk Ch. 2, §2.3 "Balance Sheet Analysis" (market-to-book, debt-equity, Eq. 2.5), pp. 30–33.

5. True or false on ratios — Higgins 2.3, p. 71

What is asked. For each of seven claims about ratios, say whether it is true or false and give the reason. (a) assets/equity versus one plus liabilities/equity; (b) whether ROE is always at least ROA; (c) whether the collection period must be shorter than the payables period; (d) whether the current ratio is always at least the acid-test ratio; (e) whether a higher asset turnover is preferable, all else equal; (f) whether equal earnings yields allow different P/E ratios; (g) whether paper gains are worth less than realized gains when there are no taxes or costs.

Approach. Write each ratio from its definition and see whether the claim follows algebraically (an identity), depends on a sign condition, or is an empirical tendency that need not hold. One counterexample is enough to make an "always" claim false.

Notation.

- A, L, E: assets, liabilities, equity (\$; $L \geq 0$, $E > 0$ unless stated).
- NI: net income (any sign); S: sales; CA, CL, Inv: current assets, current liabilities, inventory.
- P: share price; EPS: earnings per share.

Step 1 — (a) Assets-to-equity. Substitute $A = L + E$ and split the fraction:

$$\begin{aligned} \frac{A}{E} &= \frac{L+E}{E} \\ &= \frac{L}{E} + \frac{E}{E} \\ &= 1 + \frac{L}{E}. \end{aligned} \tag{16}$$

Answer (a). True: it is the balance sheet identity divided by E.

Step 2 — (b) ROE versus ROA. Multiply and divide ROE by A to relate it to ROA:

$$\begin{aligned} ROE &= \frac{NI}{E} \\ &= \frac{NI}{A} \cdot \frac{A}{E} \\ &= ROA \times \frac{A}{E}. \end{aligned} \tag{17}$$

By (16), $A/E = 1 + L/E \geq 1$ when $L \geq 0$. If $NI \geq 0$, multiplying $ROA \geq 0$ by a number ≥ 1 gives $ROE \geq ROA$. If $NI < 0$, the same multiplier makes ROE *more* negative. Counterexample: $NI = -10$, $A = 100$, $E = 50$ (so $L = 50$):

$$\begin{aligned} ROA &= (-10)/(100) = -0.10, \\ A/E &= (100)/(50) = 2, \\ ROE &= (-0.10)(2) = -0.20 < ROA. \end{aligned}$$

Leverage amplifies returns in both directions, so "always" fails for loss-making firms (and the comparison breaks down with negative equity).

Answer (b). False as stated (true only when $NI \geq 0$ and $E > 0$).

Step 3 — (c) Collection period versus payables period. The collection period $AR/(S/365)$ and the payables period $AP/(COGS/365)$ are set by industry practice and bargaining power with customers and suppliers. A payables period longer than the collection period is convenient (suppliers then finance the receivables), but no identity forces it.

Answer (c). False: it is a desirable situation, not a requirement.

Step 4 — (d) Current ratio versus acid test. The acid test removes inventory from the numerator, so subtract the two ratios:

$$\frac{CA}{CL} - \frac{CA - Inv}{CL} = \frac{CA - (CA - Inv)}{CL} = \frac{Inv}{CL} \geq 0,$$

since inventory is never negative and $CL > 0$.

Answer (d). True.

Step 5 — (e) Asset turnover. By the DuPont decomposition, obtained by multiplying and dividing NI/E by S and by A,

$$ROE = \frac{NI}{S} \cdot \frac{S}{A} \cdot \frac{A}{E},$$

a higher asset turnover S/A, with margin and leverage held fixed, raises ROE: the same sales are produced with fewer assets.

Answer (e). True (all else equal).

Step 6 — (f) Earnings yield and P/E. The earnings yield is the reciprocal of the P/E ratio, $EPS/P = 1/(P/EPS)$. If two firms have the same EPS/P, taking reciprocals gives the same P/EPS.

Answer (f). False: equal earnings yields imply equal P/E ratios.

Step 7 — (g) Paper gains. Without taxes and transaction costs, a paper gain can be turned into cash at any moment by selling the asset at its market value, so it is worth exactly as much as an equal realized gain.

Answer (g). False.

Intuition. These are accounting identities, not empirical regularities: (a), (d), (f) follow from definitions; (b) holds only because leverage is a multiplier, which also magnifies losses; (g) is the economist's view that value is market value, whether or not the accountant has recognized it.

Common mistakes.

- Answering (b) True without noticing that the multiplier A/E also scales losses.
- Confusing earnings yield with dividend yield in (f).
- Treating "should" in (c) as an accounting rule rather than a matter of bargaining.

Book answer. Matches the book's answers (p. 401) for (a), (c)–(g). For (b) the book answers True, arguing that ROA can exceed ROE only with negative liabilities; that argument implicitly assumes $NI \geq 0$. With a net loss, $ROE < ROA$ by (17), so I mark (b) False as stated.

Ref: Higgins Ch. 2, § "The Three Determinants of ROE", pp. 38–40; "Asset Turnover", pp. 42–48; "Financial Leverage", pp. 48–53; "The Value Problem", pp. 56–58; Ch. 1, "Economic Income vs. Accounting Income", pp. 26–27.

6. Book and market leverage of two firms — Berk 2.21, p. 55

What is asked. (a) Debt divided by market equity for each firm. (b) Debt divided by book equity for each firm. (c) How many times operating income covers interest for each firm. (d) Which firm is more likely to have trouble paying its debt.

Approach. Debt-equity ratios measure the size of the debt claim relative to the owners' stake; the coverage ratio measures the flow of earnings available to pay interest (Berk §2.5 "Leverage Ratios"). Distress is about paying the interest bill, so (d) is judged mainly by coverage, backed by leverage.

Notation.

- D: debt; BE: book equity; ME: market equity; OI: operating income; Int: interest expense (\$ million, > 0); subscripts A, B for the firm.

Assumption. Interest coverage uses operating income (here equal to EBIT), one of the earnings measures Berk lists (p. 37).

Step 1 — Market debt-equity (a). Divide debt by market equity for each firm:

$$\begin{aligned}(D/E)_A^{mkt} &= (500)/(400)\\ &= 1.25,\\ (D/E)_B^{mkt} &= (80)/(40)\\ &= 2.00.\end{aligned}$$

Answer (a). A: 1.25; B: 2.00.

Step 2 — Book debt-equity (b). Divide the same debt by book equity:

$$\begin{aligned}(D/E)_A^{bk} &= (500)/(300)\\ &= 1.667,\\ (D/E)_B^{bk} &= (80)/(35)\\ &= 2.286.\end{aligned}$$

Answer (b). A: 1.67; B: 2.29.

Step 3 — Interest coverage (c). Coverage is operating income over interest:

$$\begin{aligned}\text{Coverage} &= OI/Int, \qquad (18)\\ \text{Coverage}_A &= (100)/(50)\\ &= 2.00,\\ \text{Coverage}_B &= (8)/(7)\\ &= 1.143.\end{aligned}$$

Answer (c). A: 2.00; B: 1.14.

Step 4 — Who struggles more (d). The cushion is operating income above interest; the fall in operating income the firm can absorb before earnings no longer cover interest is that cushion divided by operating income:

$$\begin{aligned}\text{Max. fall}_A &= (OI_A - Int_A)/OI_A\\ &= (100-50)/(100)\\ &= 0.50,\\ \text{Max. fall}_B &= (OI_B - Int_B)/OI_B\\ &= (8-7)/(8)\\ &= 0.125.\end{aligned}$$

Answer (d). Firm B: higher leverage on both measures, coverage of only 1.14, and a fall of just 12.5% in operating income leaves it unable to pay interest out of earnings (Firm A can absorb 50%).

Intuition. Ratios scale away firm size. Firm B is much smaller (debt of 80 against A's 500, i.e. $80/500 = 16\%$), but it carries proportionally more debt and has almost no cushion of earnings above its interest bill. The ranking does not depend on using book or market equity here; when the two disagree, market values reflect what the claims are worth today.

Common mistakes.

- Comparing debt levels (500 vs. 80) instead of ratios.
- Inverting coverage (interest over income) and reading a low number as safe.
- Concluding from book ratios alone when market equity is available.

Ref: Berk Ch. 2, §2.3 (debt-equity ratios), pp. 30–33; §2.5 "Leverage Ratios", p. 37.

7. DuPont for a retail firm — Berk 2.27, p. 56

What is asked. (a) The firm's return on equity now. (b) The ROE if the net margin rises to 4%, all else fixed. (c) The ROE if, in addition, revenue rises 20% with assets and liabilities unchanged.

Approach. The DuPont identity (Berk Eq. 2.18) writes ROE as margin times asset turnover times the equity multiplier. Each scenario changes one or two factors, so ROE is recomputed by substitution; a levels check (sales, net income, equity) confirms each result.

Notation.

- m: net profit margin NI/S; T: total asset turnover S/A; $A = 44$: total assets; $BE = 18$: book equity (\$ million); $EM = A/BE$: equity multiplier.

Assumption. In (c) book equity stays at 18 (the statement fixes assets and liabilities), so only T and m change relative to (a).

Step 1 — Equity multiplier and DuPont identity. The identity follows from multiplying and dividing NI/BE by S and A; the multiplier comes from the data:

$$ROE = \frac{NI}{BE} = \frac{NI}{S} \cdot \frac{S}{A} \cdot \frac{A}{BE} = m \times T \times EM, \tag{19}$$

$$\begin{aligned} EM &= (44)/(18) \\ &= 2.4444. \end{aligned} \tag{20}$$

Step 2 — Current ROE (a). Substitute $m = 0.035$, $T = 1.8$ and (20) into (19):

$$\begin{aligned} ROE_a &= (0.035)(1.8)(2.4444) \\ &= (0.063)(2.4444) \\ &= 0.1540 \\ &= 15.4\%. \end{aligned}$$

Check in levels:

$$\begin{aligned} S &= T \times A = (1.8)(44) = 79.2, \\ NI &= m \times S = (0.035)(79.2) = 2.772, \\ ROE_a &= NI/BE = (2.772)/(18) = 0.154. \end{aligned}$$

Answer (a). $ROE = 15.4\%$.

Step 3 — Higher margin (b). Only m changes, to 0.04:

$$\begin{aligned} ROE_b &= (0.04)(1.8)(2.4444) \\ &= (0.072)(2.4444) \\ &= 0.1760 \\ &= 17.6\%. \end{aligned}$$

Answer (b). $ROE = 17.6\%$.

Step 4 — Higher margin and 20% more revenue (c). With A fixed, revenue up 20% means turnover up 20%:

$$\begin{aligned} T_c &= (1.8)(1.20) \\ &= 2.16, \\ ROE_c &= (0.04)(2.16)(2.4444) \\ &= (0.0864)(2.4444) \\ &= 0.2112 \\ &= 21.1\%. \end{aligned} \tag{21}$$

In levels: $S = (79.2)(1.20) = 95.04$, $NI = (0.04)(95.04) = 3.8016$, $ROE = (3.8016)/(18) = 0.2112$.

Answer (c). $ROE = 21.1\%$.

Intuition. ROE is a product, so proportional changes multiply: +14.3% in margin ($0.04/0.035 = 1.143$) and +20% in turnover give $(1.143)(1.2) = 1.371$ times the original ROE, and indeed $0.2112/0.1540 = 1.371$. Leverage is unchanged throughout, so all the gain comes from operations.

Common mistakes.

- Adding percentage changes ($14.3\% + 20\%$) instead of multiplying the factors.
- Using ROA ($m \times T$) as the answer and forgetting the equity multiplier.
- In (c), raising assets with revenue: the statement holds assets fixed, so turnover rises.

Ref: Berk Ch. 2, §2.5 "Investment Returns" and "The DuPont Identity" (Eq. 2.18), pp. 37–39.

8. Atlantic vs. Pacific: ROE, ROA, ROIC — Higgins 2.7, p. 73

What is asked. (a) Return on equity, on assets and on invested capital for each company. (b) Why Pacific's ROE is much higher, and whether that makes it the better company. (c) Why Atlantic's ROA is higher, and what that says. (d) How the ROICs compare and what that says.

Approach. Compute net income after interest and tax, then the three ratios from Higgins' definitions. To explain the gaps, derive the leverage identity linking ROE to ROIC: it separates the return of the business (ROIC) from the boost that debt gives shareholders when ROIC exceeds the after-tax cost of debt.

Notation.

- $EBIT$: earnings before interest and taxes; D: debt; E: equity ($ million); $r = 0.07$: interest rate; $t = 0.30$: tax rate.
- $A = D + E$: total assets (no other liabilities); NI: net income.
- $ROE = NI/E$; $ROA = NI/A$; $ROIC = EBIT(1-t)/(D+E)$ (Higgins' definitions).

Step 1 — Net income. Subtract interest rD from EBIT, then apply the tax:

$$\begin{aligned} NI &= (EBIT - rD)(1-t), \\ NI_{Atl} &= (450 - (0.07)(290))(0.70) \\ &= (450 - 20.3)(0.70) \\ &= (429.7)(0.70) \\ &= 300.79, \\ NI_{Pac} &= (470 - (0.07)(1490))(0.70) \\ &= (470 - 104.3)(0.70) \\ &= (365.7)(0.70) \\ &= 255.99. \end{aligned} \tag{22}$$

Step 2 — ROE, ROA, ROIC (a). Total assets are $A_{Atl} = 290 + 910 = 1200$ and $A_{Pac} = 1490 + 370 = 1860$. After-tax EBIT is $(450)(0.70) = 315$ for Atlantic and $(470)(0.70) = 329$ for Pacific. With (22):

$$\begin{aligned} ROE_{Atl} &= (300.79)/(910) \\ &= 0.3305, \\ ROA_{Atl} &= (300.79)/(1200) \\ &= 0.2507, \\ ROIC_{Atl} &= (315)/(1200) \\ &= 0.2625, \\ ROE_{Pac} &= (255.99)/(370) \\ &= 0.6919, \\ ROA_{Pac} &= (255.99)/(1860) \\ &= 0.1376, \\ ROIC_{Pac} &= (329)/(1860) \\ &= 0.1769. \end{aligned}$$

Answer (a). Atlantic: ROE 33.1%, ROA 25.1%, ROIC 26.3%. Pacific: ROE 69.2%, ROA 13.8%, ROIC 17.7%.

Step 3 — Why Pacific's ROE is higher (b). Derive ROE in terms of ROIC. Expand (22), replace $EBIT(1-t)$ by $ROIC\,(D+E)$ (the definition of ROIC), collect the D terms and divide by E:

$$\begin{aligned} NI &= EBIT(1-t) - r(1-t)D \\ &= ROIC\,(D+E) - r(1-t)D \\ &= ROIC\,E + \big(ROIC - r(1-t)\big)D, \\ ROE = \frac{NI}{E} &= ROIC + \big(ROIC - r(1-t)\big)\frac{D}{E}. \end{aligned} \tag{23}$$

The after-tax cost of debt is $r(1-t) = (0.07)(0.70) = 0.049$. Substituting each company's numbers into (23):

$$\begin{aligned} ROE_{Atl} &= (0.2625) + (0.2625 - 0.049)(290/910) \\ &= (0.2625) + (0.2135)(0.31868) \\ &= (0.2625) + (0.06804) \\ &= 0.3305, \\ ROE_{Pac} &= (0.176882) + (0.176882 - 0.049)(1490/370) \\ &= (0.176882) + (0.127882)(4.02703) \\ &= (0.176882) + (0.51498) \\ &= 0.6919, \end{aligned}$$

reproducing Step 2. Pacific's ROE comes from its debt-equity ratio of 4.03 (vs. 0.32), not from better operations: its leverage term is 0.515 against Atlantic's 0.068, while its ROIC is lower.

Answer (b). Pacific's higher ROE is a leverage effect ($D/E = 4.03$ vs. 0.32); it does not make Pacific a better company, and its shareholders bear much more risk.

Step 4 — Why Atlantic's ROA is higher (c). ROA puts the income of shareholders only (after interest) over the capital of both shareholders and creditors, so it penalizes leverage: Pacific's interest bill of 104.3 (vs. 20.3) lowers the numerator while the debt raises the denominator.

Answer (c). Atlantic's ROA is higher partly because it has less debt; ROA mixes financing with operating performance and says little on its own.

Step 5 — ROIC comparison (d). ROIC uses after-tax EBIT, which does not depend on how the firm is financed, over all invested capital. Atlantic earns 26.3% against Pacific's 17.7%. Both exceed the after-tax cost of debt (4.9%), which is why leverage raises ROE in (23); whether either beats its full cost of capital depends on business risk, not shown here.

Answer (d). Atlantic's ROIC is higher (26.3% vs. 17.7%): its assets are more productive, so on operating performance Atlantic is the better company.

Intuition. ROE answers "how much per dollar of owners' money", ROA mixes in the financing choice, and ROIC isolates the business. Compare operating performance with ROIC, then read ROE as ROIC plus a leverage bonus (or penalty): the bonus is the spread $ROIC - r(1-t)$ times D/E, and it turns negative, magnifying losses, if ROIC falls below the after-tax cost of debt.

Common mistakes.

- Computing ROIC with net income instead of after-tax EBIT, which reintroduces the financing choice.
- Forgetting that interest is tax-deductible: the cost of debt in (23) is $r(1-t)$, not r.
- Ranking companies by ROE without adjusting for leverage and risk.

Book answer. Matches the book's answer (pp. 402–403): ROE 33.1% / 69.2%, ROA 25.1% / 13.8%, ROIC 26.3% / 17.7%; same conclusions for (b)–(d).

Ref: Higgins Ch. 2, § "The Three Determinants of ROE", pp. 38–40; "The Risk Problem" (ROA, ROIC), pp. 54–56.

9. Nave Espacial Jetson: cash flows — Ross 2.14, p. 48

What is asked. (a) The cash generated by operations in 2009 (fluxo de caixa operacional). (b) The net cash paid to creditors. (c) The net cash paid to shareholders. (d) Given that net fixed assets rose by 27,000, the investment in net working capital (acréscimo ao CCL).

Approach. Ross's cash flow identity: cash flow from assets equals cash flow to creditors plus cash flow to shareholders, and cash flow from assets equals operating cash flow minus net capital spending minus the change in net working capital. Parts (a)–(c) are direct; (d) is the one unknown left in the identity, solved by rearranging it.

Notation.

- S: sales; C: costs; X: other expenses; Dep: depreciation; Int: interest; Tax: taxes; Div: dividends ($).
- OCF: operating cash flow (fluxo de caixa operacional); NCS: net capital spending; ΔNWC: change in net working capital (CCL); CFA: cash flow from assets; CFC, CFS: cash flow to creditors and to shareholders.

Step 1 — EBIT and operating cash flow (a). EBIT subtracts operating costs from sales; OCF adds back depreciation (not a cash cost) and subtracts the taxes actually paid (Ross §2.4):

$$\begin{aligned} EBIT &= S - C - X - Dep \\ &= (196{,}000) - (104{,}000) - (6{,}800) - (9{,}100) \\ &= 76{,}100, \end{aligned} \tag{24}$$

$$OCF = EBIT + Dep - Tax \tag{25}$$

$$\begin{aligned} &= (76{,}100) + (9{,}100) - (21{,}455) \\ &= 63{,}745. \end{aligned} \tag{26}$$

Answer (a). $OCF = \$63{,}745$.

Step 2 — Cash flow to creditors (b). Creditors received the interest and, in addition, the 7,300 of debt that was retired; net new long-term borrowing is therefore −7,300:

$$\begin{aligned} CFC &= Int - \text{net new borrowing} && (27)\\ &= (14{,}800) - (-7{,}300) \\ &= 22{,}100. && (28)\end{aligned}$$

Answer (b). $CFC = \$22{,}100$.

Step 3 — Cash flow to shareholders (c). Shareholders received the dividends but put 5,700 of new equity in:

$$\begin{aligned} CFS &= Div - \text{net new equity} \\ &= (10{,}400) - (5{,}700) \\ &= 4{,}700. && (29)\end{aligned}$$

Answer (c). $CFS = \$4{,}700$.

Step 4 — Change in net working capital (d). Cash flow from assets is the sum of what went to the two groups of investors, (28) and (29). Net capital spending is the rise in net fixed assets plus the depreciation that was charged against them. Then rearrange $CFA = OCF - NCS - \Delta NWC$ for the unknown:

$$\begin{aligned} CFA &= CFC + CFS \\ &= (22{,}100) + (4{,}700) \\ &= 26{,}800, && (30)\\ NCS &= \Delta\text{NFA} + Dep \\ &= (27{,}000) + (9{,}100) \\ &= 36{,}100, && (31)\\ CFA &= OCF - NCS - \Delta NWC \\ \Rightarrow\ \Delta NWC &= OCF - NCS - CFA \\ &= (63{,}745) - (36{,}100) - (26{,}800) \\ &= 845.\end{aligned}$$

Answer (d). The addition to net working capital was \$845.

Intuition. The firm's operations threw off 63,745; it reinvested 36,100 in fixed assets and 845 in working capital, and paid the remaining 26,800 to its investors, mostly to creditors (interest plus debt repayment). The identity is just "cash generated = cash reinvested + cash paid out".

Common mistakes.

- Treating retired debt as a source of cash for creditors: repayment is cash *to* them, so it is added (net new borrowing is negative).
- Using the 27,000 rise in net fixed assets as NCS: depreciation must be added back.
- Subtracting interest in OCF: in Ross's definition interest belongs to cash flow to creditors.

Ref: Ross Cap. 2, §2.4 "Fluxo de caixa", pp. 37–44.

10. Blake's Restaurant Supply: from statements to cash — Higgins 1.11, p. 34

What is asked. (a) The cash actually collected from customers in 2021. (b) The cost of the goods produced in 2021 (not of those sold). (c) Capital expenditures in 2021, with no assets sold. (d) Cash flow from operations in 2021, assuming no financing flows.

Approach. Each answer comes from a stock–flow equation: closing balance = opening balance + inflows − outflows. Write it for receivables, inventory and net fixed assets, and solve for the unknown flow. For (d), the change in cash is the sum of operating, investing and financing flows; an indirect calculation from net income is a cross-check.

Notation.

- Subscripts 0 = end-2020, 1 = end-2021; S: net sales 2021; $COGS$: cost of goods sold 2021; Dep: depreciation 2021; NI: net income 2021 ($ million).
- AR, Inv, AP, NFA, $Cash$: balance sheet accounts; $Capex$: capital expenditures; CFO: cash flow from operations.

Step 1 — Cash collected from sales (a). Receivables grow with sales and shrink with collections, $AR_1 = AR_0 + S -$ Collections; solve for collections:

$$\begin{aligned} \text{Collections} &= S - (AR_1 - AR_0) \\ &= (782) - (87 - 57) \\ &= (782) - (30) \\ &= 752. \end{aligned}$$

Answer (a). $752 million collected.

Step 2 — Cost of goods produced (b). Inventory grows with production and shrinks with goods sold, $Inv_1 = Inv_0 +$ Production $- COGS$; solve for production:

$$\begin{aligned} \text{Production} &= COGS + (Inv_1 - Inv_0) \\ &= (502) + (29 - 39) \\ &= (502) + (-10) \\ &= 492. \end{aligned}$$

Answer (b). Cost of goods produced = $492 million.

Step 3 — Capital expenditures (c). Net fixed assets grow with capex and shrink with depreciation (no disposals), $NFA_1 = NFA_0 + Capex - Dep$; solve for capex:

$$\begin{aligned} Capex &= (NFA_1 - NFA_0) + Dep \\ &= (482 - 404) + (61) \\ &= (78) + (61) \\ &= 139. \end{aligned} \tag{32}$$

Answer (c). Capital expenditures = $139 million.

Step 4 — Cash flow from operations (d). With no financing flows, the change in cash is operating plus investing cash flow, and investing is $-Capex$ from (32); solve for CFO:

$$\begin{aligned} \Delta Cash &= CFO - Capex, \\ CFO &= \Delta Cash + Capex \\ &= (135 - 86) + (139) \\ &= (49) + (139) \\ &= 188. \end{aligned} \tag{33}$$

Indirect check: start from net income, add back depreciation, subtract increases in operating assets and add increases in operating liabilities, with $\Delta AR = 87 - 57 = 30$, $\Delta Inv = 29 - 39 = -10$, $\Delta AP = 44 - 39 = 5$:

$$\begin{aligned} CFO &= NI + Dep - \Delta AR - \Delta Inv + \Delta AP \\ &= (142) + (61) - (30) - (-10) + (5) \\ &= 188. \end{aligned}$$

Answer (d). Cash flow from operations = \$188 million.

Intuition. Accrual accounting records sales when made and costs when goods are sold; cash records them when money moves. Rising receivables keep 30 of sales out of the bank; running down inventory and stretching payables put cash back in. That is why CFO (188) exceeds net income (142).

Common mistakes.

- Adding the rise in receivables to sales in (a): uncollected sales reduce cash, they do not increase it.
- Confusing goods produced with goods sold in (b): production differs from COGS by the change in inventory.
- Forgetting to add depreciation back in (c): the change in net fixed assets is capex net of depreciation.

Book answer. Matches the book's answer (pp. 399–400): 752, 492, 139, 188 (\$ million).

Ref: Higgins Ch. 1, § "Sources and Uses Statements", pp. 16–18; "The Cash Flow Statement", pp. 18–22.

11. Indústrias Dahlia: cash flow identity — Ross 2.21, p. 49

What is asked. (a) Net income for 2009. (b) Operating cash flow for 2009. (c) Cash flow from assets for 2009, and whether a negative value is possible. (d) With no new debt, the cash flows to creditors and to shareholders, and an interpretation of the signs of all answers (a)–(d).

Approach. Build the income statement to get net income and taxes, apply Ross's definitions of OCF, NCS and ΔNWC to get cash flow from assets, then split it between creditors and shareholders with the identity $CFA = CFC + CFS$. The new equity raised is the residual that makes the identity hold.

Notation.

- S, $COGS$, Dep, Int, Div: sales, cost of goods sold, depreciation, interest, dividends (\$); $\tau = 0.34$.
- NFA, CA, CL: net fixed assets, current assets, current liabilities; subscripts 0 (start of year), 1 (end of year). Other symbols as in Exercise 9.

Step 1 — Net income (a). Go down the income statement: EBIT, then pretax income after

interest, then tax at 34%:

$$\begin{aligned} EBIT &= S - COGS - Dep \\ &= (22{,}800) - (16{,}050) - (4{,}050) \\ &= 2{,}700, \end{aligned} \tag{34}$$

$$\begin{aligned} EBT &= EBIT - Int \\ &= (2{,}700) - (1{,}830) \\ &= 870, \\ Tax &= \tau\, EBT \\ &= (0.34)(870) \\ &= 295.80, \end{aligned} \tag{35}$$

$$\begin{aligned} NI &= EBT - Tax \\ &= (870) - (295.80) \\ &= 574.20. \end{aligned}$$

Answer (a). Net income = \$574.20.

Step 2 — Operating cash flow (b). Apply (25) with (34) and (35):

$$\begin{aligned} OCF &= EBIT + Dep - Tax \\ &= (2{,}700) + (4{,}050) - (295.80) \\ &= 6{,}454.20. \end{aligned} \tag{36}$$

Answer (b). OCF = \$6,454.20.

Step 3 — Cash flow from assets (c). Net capital spending is the change in net fixed assets plus depreciation; the change in net working capital is closing minus opening $CA - CL$; then subtract both from OCF:

$$\begin{aligned} NCS &= (NFA_1 - NFA_0) + Dep \\ &= (16{,}800 - 13{,}650) + (4{,}050) \\ &= (3{,}150) + (4{,}050) \\ &= 7{,}200, \end{aligned} \tag{37}$$

$$\begin{aligned} \Delta NWC &= (CA_1 - CL_1) - (CA_0 - CL_0) \\ &= (5{,}930 - 3{,}150) - (4{,}800 - 2{,}700) \\ &= (2{,}780) - (2{,}100) \\ &= 680, \end{aligned} \tag{38}$$

$$\begin{aligned} CFA &= OCF - NCS - \Delta NWC \\ &= (6{,}454.20) - (7{,}200) - (680) \\ &= -1{,}425.80. \end{aligned} \tag{39}$$

A negative CFA is possible: the firm invested $NCS + \Delta NWC = 7{,}200 + 680 = 7{,}880$ in fixed assets and working capital, more than the 6,454.20 its operations generated, and raised the difference $7{,}880 - 6{,}454.20 = 1{,}425.80$ from investors.

Answer (c). $CFA = -$\$1,425.80; yes, possible, because investment exceeded operating cash flow.

Step 4 — Cash flow to creditors and shareholders (d). With no new debt, (27) leaves only interest; the shareholders' flow is what remains of CFA; and the equity raised follows from

$CFS = Div -$ net new equity:

$$\begin{aligned} CFC &= Int - \text{net new borrowing} \\ &= (1{,}830) - (0) \\ &= 1{,}830, \end{aligned} \tag{40}$$

$$\begin{aligned} CFS &= CFA - CFC \\ &= (-1{,}425.80) - (1{,}830) \\ &= -3{,}255.80, \end{aligned} \tag{41}$$

$$\begin{aligned} \text{net new equity} &= Div - CFS \\ &= (1{,}300) - (-3{,}255.80) \\ &= 4{,}555.80. \end{aligned}$$

Answer (d). $CFC = \$1{,}830$; $CFS = -\$3{,}255.80$ (shareholders contributed \$4,555.80 of new equity, net of the \$1,300 dividend).

Intuition. Signs: net income and OCF are positive (the business is profitable and generates cash); CFA is negative (growth spending exceeds that cash); CFC is positive (creditors received their interest); CFS is negative (shareholders put in 4,555.80 of new equity, net of the 1,300 dividend they received). A growing firm is a net user of investors' cash.

Common mistakes.

- Reading a negative CFA as a loss: it reflects investment, not unprofitability (NI is positive).
- Computing ΔNWC from current assets alone, forgetting the rise in current liabilities.
- Setting CFS equal to dividends: new equity raised must be netted out.

Ref: Ross Cap. 2, §2.2 "A demonstração de resultados do exercício", pp. 30–34; §2.4 "Fluxo de caixa", pp. 37–44.

12. Global's marketing campaign — Berk *2.19, p. 54

What is asked. (a) Global's EBIT in the campaign scenario (sales up 15%, operating margin 4.50%). (b) Its net income, with interest unchanged and the 2009 tax ratio. (c) Its share price if the P/E ratio and the share count stay as reported.

Approach. Propagate the shock down the income statement of Berk's Table 2.3: new sales times new margin gives EBIT; subtracting the fixed interest gives pretax income; the effective tax ratio gives net income. With P/E constant, price is P/E times the new EPS (Eqs. 2.8 and 2.19). Fixed interest makes this an exercise in financial leverage.

Notation.

- S: net sales; m: operating margin; $EBIT$; Int: interest expense; τ: taxes / pretax income; NI: net income (\$ million); N: shares (million); P: share price (\$).

Data used (Berk Table 2.3, p. 34; Eq. 2.8, p. 34; p. 39). Reported 2010: net sales 186.7, operating income = EBIT = 10.4 (no other income), so $m = 10.4/186.7 = 5.57\%$; interest expense 7.7; pretax income 2.7; taxes 0.7; net income 2.0; $N = 3.6$ million shares; P/E $= 36/2.0 = 18$ (p. 39), so the reported price is $P = (18)(2.0/3.6) = 10$. Reported 2009: pretax income 2.5, taxes 0.6.

Assumption. The 5.57% margin is Global's reported 2010 margin, so the campaign scenario replaces the reported 2010: sales are 15% above the reported 186.7 and interest stays at the reported 7.7. The tax rate is the 2009 ratio, as the statement says. (If the 2010 ratio 0.7/2.7 were meant instead, the results are given at the end.)

Step 1 — EBIT (a). Raise sales by 15%, then apply the new operating margin (EBIT = operating income, no other income):

$$\begin{aligned} S' &= (186.7)(1.15) \\ &= 214.705, \end{aligned} \tag{42}$$

$$\begin{aligned} EBIT' &= m'S' \\ &= (0.045)(214.705) \\ &= 9.6617. \end{aligned} \tag{43}$$

Answer (a). $EBIT = \$9.66$ million.

Step 2 — Net income (b). Compute the 2009 tax ratio, subtract the unchanged interest from (43), then apply the tax:

$$\begin{aligned} \tau_{2009} &= (0.6)/(2.5) \\ &= 0.24, \end{aligned} \tag{44}$$

$$\begin{aligned} EBT' &= EBIT' - Int \\ &= (9.6617) - (7.7) \\ &= 1.9617, \\ NI' &= EBT'(1 - \tau_{2009}) \\ &= (1.9617)(1 - 0.24) \\ &= (1.9617)(0.76) \\ &= 1.4909. \end{aligned} \tag{45}$$

Answer (b). Net income = \$1.49 million.

Step 3 — Share price (c). From Eq. 2.19, $P = (P/E) \times EPS$, and from Eq. 2.8, $EPS = NI/N$; with P/E and N unchanged,

$$\begin{aligned} P' &= (P/E) \times NI'/N \\ &= (18)(1.4909/3.6) \\ &= (18)(0.41414) \\ &= 7.45. \end{aligned} \tag{46}$$

Answer (c). Share price $\approx$ \$7.45 (down from \$10).

Alternative tax rate. With the 2010 ratio $\tau = 0.7/2.7 = 0.25926$:

$$\begin{aligned} NI' &= (1.9617)(1 - 0.25926) \\ &= (1.9617)(0.74074) \\ &= 1.4531, \\ P' &= (18)(1.4531/3.6) \\ &= 7.27. \end{aligned}$$

Intuition. Sales up 15% but margin down by about a fifth ($1 - 4.50/5.57 = 0.19$) leaves EBIT below the reported 10.4: $9.6617/10.4 - 1 = -7.1\%$. Because interest is fixed, that fall in EBIT becomes a fall of $1.9617/2.7 - 1 = -27\%$ in pretax income: that is financial leverage at work, and with a constant P/E the price falls in proportion to earnings, from \$10 to about \$7.45.

Common mistakes.

- Growing 2009 sales (176.1) instead of the reported 2010 sales whose margin was 5.57%.

- Letting interest scale with sales: the statement fixes it at 7.7.
- Using a statutory tax rate instead of the 2009 ratio of taxes to pretax income (0.6/2.5).

Ref: Berk Ch. 2, §2.4, pp. 33–35 (Table 2.3); §2.5 "Profitability Ratios" and "Valuation Ratios" (Eqs. 2.10, 2.19), pp. 35–36, 39.

13. Quisco: develop in house or acquire with stock — Berk *2.23, p. 55

What is asked. (a) The change in EPS if Quisco spends \$500 million on in-house R&D, expensed this year, at a 35% tax rate. (b) The change in EPS if it instead pays \$900 million in its own shares for a firm with no income of its own. (c) Which route hurts reported earnings less, and whether it is also the cheaper one.

Approach. EPS is net income over shares (Berk Eq. 2.8). The R&D route changes the numerator (by the after-tax expense); the stock route changes only the denominator (new shares). Whether a route is cheaper is a question about value given up by existing shareholders, not about EPS.

Notation.

- $N = 6{,}500$ million shares; $P = 18$ (\$); $EPS_0 = 0.80$: EPS without the technology.
- $R = 500$: in-house R&D cost; $V = 900$: value of shares paid for the target (\$ million); $\tau = 0.35$; ΔN: new shares issued (million).

Assumption. New shares are issued at the current price of \$18.

Step 1 — Baseline net income. Invert $EPS = NI/N$:

$$\begin{aligned} NI_0 &= EPS_0 \times N \\ &= (0.80)(6{,}500) \\ &= 5{,}200. \end{aligned} \tag{47}$$

Step 2 — In-house development (a). The R&D expense lowers pretax income by R and taxes by τR, so net income falls by $R(1-\tau)$; the share count is unchanged:

$$\begin{aligned} \Delta NI &= -R(1-\tau) \\ &= -(500)(1-0.35) \\ &= -(500)(0.65) \\ &= -325, \\ \Delta EPS_a &= \Delta NI/N \\ &= (-325)/(6{,}500) \\ &= -0.05, \\ EPS_a &= (0.80) + (-0.05) \\ &= 0.75. \end{aligned} \tag{48}$$

Answer (a). EPS falls by \$0.05, from \$0.80 to \$0.75.

Step 3 — Acquisition with stock (b). The number of new shares is the payment divided by

the price; net income is unchanged by assumption, so EPS is NI_0 over the larger share count:

$$\begin{aligned} \Delta N &= V/P \\ &= (900)/(18) \\ &= 50, \\ EPS_b &= NI_0/(N+\Delta N) \\ &= (5{,}200)/(6{,}500+50) \\ &= (5{,}200)/(6{,}550) \\ &= 0.7939, \\ \Delta EPS_b &= (0.7939)-(0.80) \\ &= -0.0061. \end{aligned} \tag{49}$$

Equivalently, $EPS_b = EPS_0 \times N/(N+\Delta N) = (0.80)(6{,}500/6{,}550) = (0.80)(0.99237) = 0.7939$: the share count rises by $50/6{,}550 = 0.76\%$ of the new total.

Answer (b). EPS falls by about \$0.006, to \$0.7939.

Step 4 — Which is cheaper (c). Compare the value each route costs existing shareholders. Stock: they give up 50 million shares worth $(50)(18) = 900$ million to the target's owners. In house: the after-tax cost is 325 million (48) (500 million pretax). The difference is $900 - 325 = 575$ million.

Answer (c). The acquisition has the smaller effect on EPS (-0.6 cents vs. -5 cents), but it is not cheaper: building in house costs \$575 million less after tax, assuming both routes deliver the same technology.

Intuition. EPS measures how accounting allocates a cost, not the cost itself. Expensing R&D hits earnings at once; paying with stock spreads the same economic transfer as dilution, which EPS records only as a small rise in the share count. A manager judged on EPS is pushed toward the more expensive route.

Common mistakes.

- Ignoring the tax deduction in (a) and answering $-500/6{,}500 = -0.077$.
- Concluding that the route with the smaller EPS hit is cheaper: the cost is the value given up, \$900 million in shares.
- Forgetting that the share count is in billions (6.5 billion = 6,500 million) when dividing.

Ref: Berk Ch. 2, §2.4 (EPS, Eq. 2.8), pp. 33–35; §2.8 "Financial Reporting in Practice", pp. 46–48.

14. Complete the balance sheet — Higgins 2.11, p. 74

What is asked. Fill every blank line of the balance sheet (receivables, total current assets, net fixed assets, payables, short-term debt, total current liabilities, long-term debt, equity, total liabilities and equity) from cash, inventory, total assets and six ratios.

Approach. Each ratio links a stock to a flow. Start from the one ratio whose stock is given and whose flow is unknown (days' sales in cash pins down sales), then use sales and inventory to get the other stocks one at a time; the balance sheet identity closes the remaining gaps.

Notation.

- S: annual sales (all on credit); $COGS$: cost of goods sold (\$); 365-day year.
- $Cash$, AR, Inv, CA, NFA, TA: cash, receivables, inventory, current assets, net fixed assets, total assets; AP, STD, CL, LTD, TL, E: payables, short-term debt, current liabilities, long-term debt, total liabilities, equity.

- Ratios: days' sales in cash = $Cash/(S/365)$; collection period = $AR/(S/365)$; inventory turnover = $COGS/Inv$; payables period = $AP/(COGS/365)$; current ratio = CA/CL; liabilities to assets = TL/TA.

Step 1 — Sales from days' sales in cash. Days' sales in cash = $Cash/(S/365) = 34$; multiply both sides by $S/365$ and solve for S, then record daily sales for the next steps:

$$\begin{aligned} S &= 365 \times Cash/34 \\ &= (365)(1{,}100{,}000)/34 \\ &= (401{,}500{,}000)/34 \\ &= 11{,}808{,}824, \qquad (50) \\ S/365 &= (1{,}100{,}000)/34 \\ &= 32{,}352.94. \end{aligned}$$

Step 2 — Receivables, current assets, net fixed assets. Receivables are 71 days of sales (50); current assets add the three items; net fixed assets are the rest of total assets:

$$\begin{aligned} AR &= 71 \times S/365 \\ &= 71 \times (32{,}352.94) \\ &= 2{,}297{,}059, \qquad (51) \\ CA &= Cash + AR + Inv \\ &= (1{,}100{,}000) + (2{,}297{,}059) + (1{,}900{,}000) \\ &= 5{,}297{,}059, \qquad (52) \\ NFA &= TA - CA \\ &= (8{,}000{,}000) - (5{,}297{,}059) \\ &= 2{,}702{,}941. \end{aligned}$$

Step 3 — COGS and payables. Inventory turnover gives COGS; payables are 36 days of COGS:

$$\begin{aligned} COGS &= 5 \times Inv \\ &= 5 \times (1{,}900{,}000) \\ &= 9{,}500{,}000, \qquad (53) \\ COGS/365 &= (9{,}500{,}000)/365 \\ &= 26{,}027.40, \\ AP &= 36 \times COGS/365 \\ &= 36 \times (26{,}027.40) \\ &= 936{,}986. \qquad (54) \end{aligned}$$

Step 4 — Current liabilities and short-term debt. The current ratio gives current liabilities from (52); short-term debt is what is left after payables (54):

$$\begin{aligned} CL &= CA/2.6 \\ &= (5{,}297{,}059)/2.6 \\ &= 2{,}037{,}330, \qquad (55) \\ STD &= CL - AP \\ &= (2{,}037{,}330) - (936{,}986) \\ &= 1{,}100{,}344. \end{aligned}$$

Step 5 — Total liabilities, long-term debt, equity. Liabilities are 75% of assets; long-term debt is the non-current part; equity is the residual:

$$\begin{aligned} TL &= (0.75)(8{,}000{,}000) \\ &= 6{,}000{,}000, \\ LTD &= TL - CL \\ &= (6{,}000{,}000) - (2{,}037{,}330) \\ &= 3{,}962{,}670, \\ E &= TA - TL \\ &= (8{,}000{,}000) - (6{,}000{,}000) \\ &= 2{,}000{,}000. \end{aligned}$$

Answer (balance sheet). The completed statement (dollars, rounded to the dollar):

Assets	$	Liabilities and equity	$
Cash	1,100,000	Accounts payable	936,986
Accounts receivable	2,297,059	Short-term debt	1,100,344
Inventory	1,900,000	Total current liabilities	2,037,330
Total current assets	5,297,059	Long-term debt	3,962,670
Net fixed assets	2,702,941	Shareholders' equity	2,000,000
Total assets	8,000,000	Total liabilities and equity	8,000,000

Intuition. Each turnover ratio converts a flow (sales, COGS) into the stock it finances; once one anchor (cash) pins down sales, the ratios fill in the balance sheet line by line, and the identity $TA = TL + E$ closes it.

Common mistakes.

- Basing the payables period on sales: the statement says it is based on COGS.
- Applying inventory turnover to sales instead of COGS.
- Taking all liabilities as current: long-term debt is $TL - CL$.

Book answer. Matches the book's answer (pp. 403–404), line by line (figures rounded to the dollar).

Ref: Higgins Ch. 2, § "Asset Turnover" (collection period, inventory turnover, payables period, days' sales in cash), pp. 42–48; "Financial Leverage" (current ratio, liabilities to assets), pp. 48–53.

15. Taco Mel: statements and cash flows for two years — Ross 2.25–2.26, pp. 50–51

What is asked. Problem 25: build an income statement and a balance sheet for 2008 and 2009 from the list of accounts. Problem 26: for 2009, compute cash flow from assets, cash flow to creditors and cash flow to shareholders.

Approach. Sort the accounts into the two statements (Ross §§2.1–2.2). Equity is not given, so it is the balancing item of each balance sheet. Then apply the cash flow identity of Ross §2.4 as in Exercises 9 and 11; net new equity is backed out from the change in equity minus the addition to retained earnings, and the check $CFA = CFC + CFS$ verifies everything.

Notation.

- Symbols as in Exercises 9 and 11; $\tau = 0.34$; RE: addition to retained earnings; NP: short-term notes (Títulos de dívida de curto prazo); E: owners' equity.

Assumption. Short-term notes are a current liability (inside NWC); Exigível a longo prazo is long-term debt; equity is the balancing item. Cash flow to creditors uses long-term debt only, following Ross §2.4.

Step 1 — Income statements (Problem 25). For each year: EBIT = sales − COGS − other expenses − depreciation; pretax income = EBIT − interest; tax at 34%; net income; and the addition to retained earnings = net income − dividends:

$$\begin{aligned} EBIT_{08} &= (7{,}233) - (2{,}487) - (591) - (1{,}038) \\ &= 3{,}117, \end{aligned} \tag{56}$$

$$\begin{aligned} EBT_{08} &= (3{,}117) - (485) \\ &= 2{,}632, \\ Tax_{08} &= (0.34)(2{,}632) \\ &= 894.88, \\ NI_{08} &= (2{,}632) - (894.88) \\ &= 1{,}737.12, \\ RE_{08} &= (1{,}737.12) - (882) \\ &= 855.12, \\ EBIT_{09} &= (8{,}085) - (2{,}942) - (515) - (1{,}085) \\ &= 3{,}543, \end{aligned} \tag{57}$$

$$\begin{aligned} EBT_{09} &= (3{,}543) - (579) \\ &= 2{,}964, \\ Tax_{09} &= (0.34)(2{,}964) \\ &= 1{,}007.76, \end{aligned} \tag{58}$$

$$\begin{aligned} NI_{09} &= (2{,}964) - (1{,}007.76) \\ &= 1{,}956.24, \\ RE_{09} &= (1{,}956.24) - (1{,}011) \\ &= 945.24. \end{aligned} \tag{59}$$

Step 2 — Balance sheets (Problem 25). Current assets are cash + receivables + inventory; total assets add net fixed assets; current liabilities are payables + short-term notes; equity is total

assets minus all liabilities:

$$\begin{aligned} CA_{08} &= (3{,}792) + (5{,}021) + (8{,}927) \\ &= 17{,}740, \\ TA_{08} &= (17{,}740) + (31{,}805) \\ &= 49{,}545, \\ CL_{08} &= (3{,}984) + (732) \\ &= 4{,}716, \\ E_{08} &= (49{,}545) - (4{,}716) - (12{,}700) \\ &= 32{,}129, \end{aligned} \tag{60}$$

$$\begin{aligned} CA_{09} &= (4{,}041) + (5{,}892) + (9{,}555) \\ &= 19{,}488, \\ TA_{09} &= (19{,}488) + (33{,}921) \\ &= 53{,}409, \\ CL_{09} &= (4{,}025) + (717) \\ &= 4{,}742, \\ E_{09} &= (53{,}409) - (4{,}742) - (15{,}435) \\ &= 33{,}232. \end{aligned} \tag{61}$$

Answer (25). The two statements:

Income statement	2008	2009	Balance sheet	2008	2009
Sales	7,233	8,085	Cash	3,792	4,041
COGS	2,487	2,942	Accounts receivable	5,021	5,892
Other expenses	591	515	Inventory	8,927	9,555
Depreciation	1,038	1,085	Current assets	17,740	19,488
EBIT	3,117	3,543	Net fixed assets	31,805	33,921
Interest	485	579	Total assets	49,545	53,409
Pretax income	2,632	2,964	Accounts payable	3,984	4,025
Taxes (34%)	894.88	1,007.76	Short-term notes	732	717
Net income	1,737.12	1,956.24	Current liabilities	4,716	4,742
Dividends	882	1,011	Long-term debt	12,700	15,435
Add. to ret. earnings	855.12	945.24	Owners' equity	32,129	33,232

Step 3 — Cash flow from assets, 2009 (Problem 26). Apply (25) with (57) and (58); net capital spending is the change in net fixed assets plus depreciation; the change in NWC uses the

current accounts of Step 2:

$$\begin{aligned} OCF &= EBIT_{09} + Dep_{09} - Tax_{09} \\ &= (3{,}543) + (1{,}085) - (1{,}007.76) \\ &= 3{,}620.24, \\ NCS &= (NFA_{09} - NFA_{08}) + Dep_{09} \\ &= (33{,}921 - 31{,}805) + (1{,}085) \\ &= (2{,}116) + (1{,}085) \\ &= 3{,}201, \\ \Delta NWC &= (CA_{09} - CL_{09}) - (CA_{08} - CL_{08}) \\ &= (19{,}488 - 4{,}742) - (17{,}740 - 4{,}716) \\ &= (14{,}746) - (13{,}024) \\ &= 1{,}722, \\ CFA &= OCF - NCS - \Delta NWC \\ &= (3{,}620.24) - (3{,}201) - (1{,}722) \\ &= -1{,}302.76. \end{aligned} \tag{62}$$

Step 4 — Cash flow to creditors and shareholders, 2009 (Problem 26). Creditors: interest minus net new long-term debt. Shareholders: the change in equity has two sources, retained earnings and new shares, so net new equity is the change in equity minus RE_{09} (59); then $CFS = Div -$ net new equity:

$$\begin{aligned} CFC &= Int - (LTD_{09} - LTD_{08}) \\ &= (579) - (15{,}435 - 12{,}700) \\ &= (579) - (2{,}735) \\ &= -2{,}156, \end{aligned} \tag{63}$$

$$\begin{aligned} \text{net new equity} &= (E_{09} - E_{08}) - RE_{09} \\ &= (33{,}232 - 32{,}129) - (945.24) \\ &= (1{,}103) - (945.24) \\ &= 157.76, \\ CFS &= Div - \text{net new equity} \\ &= (1{,}011) - (157.76) \\ &= 853.24. \end{aligned} \tag{64}$$

Check: $CFC + CFS = (-2{,}156) + (853.24) = -1{,}302.76 = CFA$ in (62).

Answer (26). $CFA = -\$1{,}302.76$; $CFC = -\$2{,}156$; $CFS = \$853.24$.

Intuition. Taco Mel's operations generated 3,620 but it spent $NCS + \Delta NWC = 3{,}201 + 1{,}722 = 4{,}923$ on fixed assets and working capital, so investors as a group financed the gap of 1,303: creditors supplied 2,156 net (2,735 of new long-term debt less 579 of interest), which covered both the gap and a net payout of 853 to shareholders.

Common mistakes.

- Putting short-term notes in long-term debt, which moves 732/717 out of NWC and into CFC.
- Treating the whole change in equity (1,103) as new equity: most of it is retained earnings.
- Reading a negative CFC as a payment by the firm: it means creditors lent more than they received.

Ref: Ross Cap. 2, §2.1 "O balanço patrimonial", pp. 25–30; §2.2, pp. 30–34; §2.4 "Fluxo de caixa", pp. 37–44.